Today

- Delta function potential
- Reflection and transmission
- The simple harmonic oscillator

2.4 Delta function potential

Consider a particle of mass m in a delta function potential well,

$$V(x) = -\alpha\delta(x), \tag{2.4.1}$$

where α is a positive constant. A schematic figure of the potential is shown below.

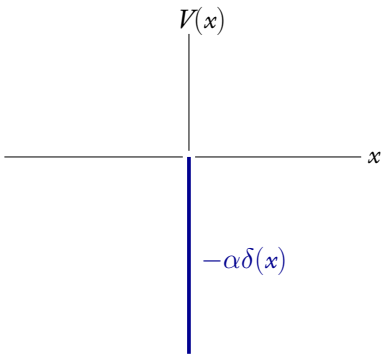


Figure 2.4.1: Delta function potential well.

To construct any bound state wavefunctions and the corresponding energies, we solve the Schrödinger equation on the left and the right sides of the potential and impose the continuity of $\psi(x)$ and the discontinuity of $\psi'(x)$ across it. Note that now that the potential has infinite discontinuity at the origin, $\psi'(x)$ is not continuous. Denoting the region $x < 0$ by I and the region $x > 0$ by II, the Schrödinger equation in region I is

$$\psi_I''(x) - \kappa^2\psi_I(x) = 0, \qquad \kappa \equiv \sqrt{-\frac{2mE}{\hbar^2}}, \tag{2.4.2}$$

where $E < 0$ for a bound state. The general solution is $\psi_I(x) = Ae^{-\kappa x} + Be^{\kappa x}$. But because the wavefunction has to vanish at $x = -\infty$, in order for it to be normalizable, we set $A = 0$, and we

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have

$$\psi_I(x) = Be^{\kappa x}. \tag{2.4.3}$$

The Schrödinger equation and the general solution in region *II* are exactly the same as in region *I*. The only difference is that the wavefunction in region *II* must vanish at $x = \infty$, which sets $B = 0$, and we have

$$\psi_{II}(x) = Ae^{-\kappa x}. \tag{2.4.4}$$

Continuity of the wavefunction across $x = 0$ imposes

$$A = B. \tag{2.4.5}$$

The discontinuity of $\psi'(x)$ across $x = 0$ follows from

$$\begin{aligned} \psi'_{II}(0) - \psi'_I(0) &= \lim_{\epsilon\to 0}\int_{-\epsilon}^{\epsilon}\psi''(x)dx = \lim_{\epsilon\to 0}\int_{-\epsilon}^{\epsilon}\frac{2m}{\hbar^2}(V(x)-E)\psi(x)dx \\ &= -\lim_{\epsilon\to 0}\frac{2m}{\hbar^2}\int_{-\epsilon}^{+\epsilon}\alpha\delta(x)\psi(x)dx = -\frac{2m}{\hbar^2}\alpha\psi(0) = -\frac{2m}{\hbar^2}\alpha A, \end{aligned} \tag{2.4.6}$$

On the other hand, taking the derivatives directly, $\psi'_{II}(0) - \psi'_I(0) = -\kappa A - \kappa B$. Therefore,

$$\kappa A + \kappa B = \frac{2m}{\hbar^2}\alpha A. \tag{2.4.7}$$

Solving (2.4.5) and (2.4.7), we have

$$\kappa = \frac{m\alpha}{\hbar^2}, \tag{2.4.8}$$

which using the definition of κ gives

$$E = -\frac{m\alpha^2}{2\hbar^2}. \tag{2.4.9}$$

Because there is only one solution, the delta function potential well supports one and only one bound state. Because $A = B$, the wavefunction decays symmetrically and exponentially on both sides. Writing $\psi_I(x) = Ae^{\kappa x}$ and $\psi_{II}(x) = Ae^{\kappa x}$, and normalizing, $1 = \int_{-\infty}^{0} A^2e^{2\kappa x} + \int_0^{\infty} A^2e^{-2\kappa x} = A^2\frac{1}{2\kappa} + A^2\frac{1}{2\kappa} = \frac{A^2}{\kappa}$, which gives $A = \sqrt{\kappa}$. Therefore, the normalized wavefunction is

$$\psi(x) = \begin{cases} \sqrt{\kappa}e^{-\kappa x} = \sqrt{\frac{m\alpha}{\hbar^2}}e^{-\frac{m\alpha}{\hbar^2}x}, & x > 0 \\ \sqrt{\kappa}e^{\kappa x} = \sqrt{\frac{m\alpha}{\hbar^2}}e^{\frac{m\alpha}{\hbar^2}x}, & x < 0. \end{cases}$$

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The energy level and the wavefunction are shown in the schematic figure below.

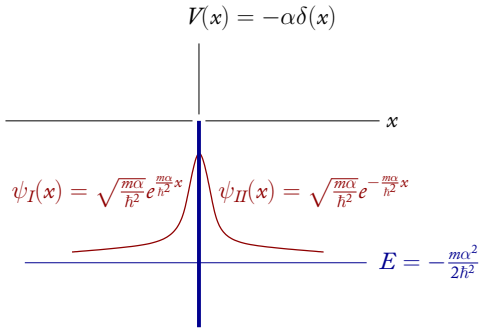


Figure 2.4.2: Energy and wavefunction of the delta function potential well.

2.5 Reflection and transmission

Now we want to consider the case of a particle of energy $E > 0$ approaching the well from the left of the finite square well potential whose bound states we have investigated. Now the particle has positive kinetic energy everywhere above the potential and it moves along from $x = -\infty$ to $x = \infty$.

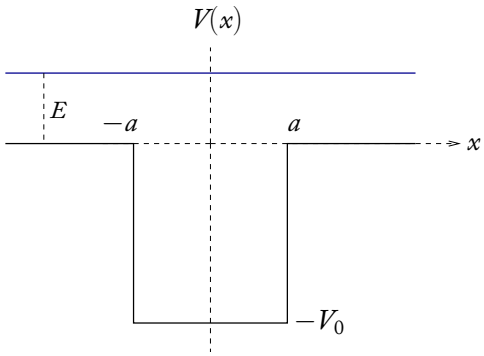


Figure 2.5.1: A particle with energy above an infinite square well.

Question 2.5.1: For a particle represented by a plane wave with energy $E > 0$ approaching the well from the left of an finite square well, (A). The wavefunction vanishes at $x = \pm\infty$. (B). The particle is partly reflected and partly transmitted at $x = -a$. (C). The particle is partly reflected and partly transmitted at $x = a$. (D). The particle is partly reflected and partly transmitted at all points it passes through. (E). The speed of the particle increases when it is between $x = -a$ and $x = a$. (F). The wavelength of the particle decreases when it is between $x = -a$ and $x = a$. (G). The particle moves along without any reflection anywhere.

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Question 2.5.2: The coefficient of transmission T should in general be defined as (A). $\frac{J_{tra}}{J_{inc}}$, (B). $\left|\frac{\psi_{tra}}{\psi_{inc}}\right|^2$, (C). Either one of the above.

Recalling that the probability current describes the particle current, the rate of particle flow though a given point, we should note that in order to compute the probability for reflection or transmission, we compare the corresponding currents. The coefficients of reflection R and transmission T are defined as

$$R = \frac{J_{ref}}{J_{inc}}, \qquad T = \frac{J_{tra}}{J_{inc}}, \tag{2.5.1}$$

where J_{inc} is the probability current for the incident wave, J_{ref} is the probability current for the reflected wave, and J_{tra} is the probability current for the transmitted wave. Observe that comparing the probability densities only does not provide a correct answer, unless the speed of the particle is the same in the corresponding regions.

Considering the finite potential well shown in the figure above, and denoting the region in $x < -a$ by I, that within $-a \leq x \leq a$ by II, and the region in $x > a$ by III, the solutions to the Schrödinger equation in each region can be written down immediately,

$$\psi_I(x) = Ae^{ikx} + Be^{-ikx}, \tag{2.5.2}$$
$$\psi_{II}(x) = C\sin\kappa x + D\cos\kappa x, \tag{2.5.3}$$
$$\psi_{III}(x) = Fe^{ikx}, \tag{2.5.4}$$

where

$$\kappa \equiv \sqrt{\frac{2mE}{\hbar^2}}, \qquad \kappa \equiv \sqrt{\frac{2m(E+V_0)}{\hbar^2}}, \tag{2.5.5}$$

A is the amplitude of the incident wave, B is the amplitude of the reflected wave, and F is the amplitude of the transmitted wave. There is no reflected wave that propagates to the left in region III. Applying the continuity of $\psi(x)$ and $\psi'(x)$ at $x = \pm a$ gives the following four equations,

$$Ae^{-ika} + Be^{ika} = -C\sin\kappa a + D\cos\kappa a, \tag{2.5.6}$$
$$ik\left(Ae^{-ika} - Be^{ika}\right) = \kappa\left(C\cos\kappa a + D\sin\kappa a\right), \tag{2.5.7}$$
$$C\sin\kappa a + D\cos\kappa a = Fe^{ika}, \tag{2.5.8}$$
$$\kappa\left(C\cos\kappa a - D\sin\kappa a\right) = ikFe^{ika} \tag{2.5.9}$$

We have four equation which we can solve for four variables B, C, D, and F in terms of the

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amplitude A of the incident wave. Our interest is in the amplitudes B and F for the reflected and the transmitted waves, and we write the solutions for B and F,

$$F = \frac{e^{-2ika}}{\cos(2\kappa a) - i\frac{k^2+\kappa^2}{2k\kappa}\sin(2\kappa a)} A \tag{2.5.10}$$

and

$$B = i\frac{\sin(2\kappa a)}{2k\kappa}(\kappa^2 - k^2)F. \tag{2.5.11}$$

The transmission coefficient is obtained using Ae^{ikx} for the incident wave and Fe^{ikx} for the transmitted wave, which gives

$$T = \frac{J_{tra}}{J_{inc}} = \left|\frac{F}{A}\right|^2 = \frac{1}{1 + \frac{V_0^2}{4E(E+V_0)}\sin^2\left(\frac{2a}{\hbar}\sqrt{2m(E+V_0)}\right)}, \tag{2.5.12}$$

and the transmission coefficient oscillates with the energy of the incident particle. Moreover,

$$R = 1 - T. \tag{2.5.13}$$

The transmission coefficient oscillates with energy as shown in the figure below.

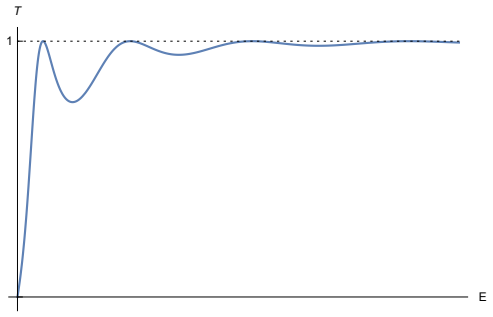
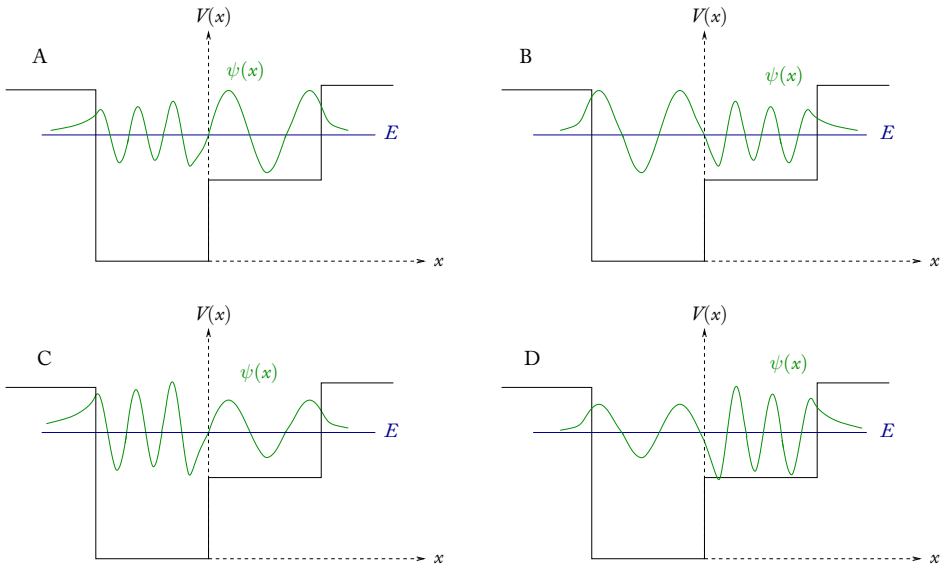


Figure 2.5.2: Transmission coefficient for a particle over a square potential well.

Question 2.5.3: Discuss why the transmission coefficient oscillates with the energy of the incident particle and what the peaks correspond to.

Question 2.5.4: Which one of the following best describes the wavefunction of a bound particle of energy E in the potential well $V(x)$?

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2.6 Step potential

Another example is a particle of energy $E > V_0$ approaching a step potential from the left in the $x < 0$ region with

$$V(x) = \begin{cases} 0, & x < 0 \\ V_0, & x > 0. \end{cases}$$

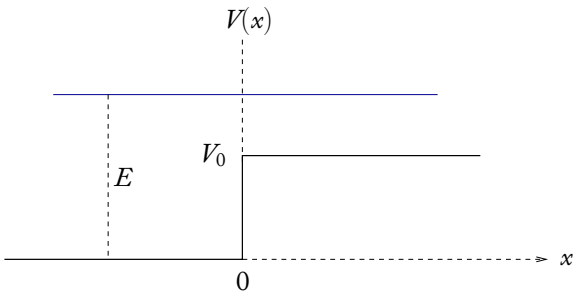


Figure 2.6.1: A particle with energy above a step potential.

Denoting the region $x < 0$ by I and that in $x > 0$ by II, and noting that there are both incident and reflected waves in region I while there is only transmitted wave in region II, we can immediately write the wavefunctions in each region,

$$\psi_I(x) = Ae^{ikx} + Be^{-ikx}, \qquad \psi_{II}(x) = Ce^{i\kappa x}, \tag{2.6.1}$$

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where

$$k = \sqrt{\frac{2mE}{\hbar^2}}, \qquad \kappa = \sqrt{\frac{2m(E-V_0)}{\hbar^2}} \tag{2.6.2}$$

Matching $\psi(x)$ and $\psi'(x)$ across $x = 0$, we have

$$A + B = C, \qquad ik(A - B) = i\kappa C. \tag{2.6.3}$$

Solving these simultaneous equations for B and C in terms of A,

$$B = \frac{k-\kappa}{k+\kappa}A, \qquad C = \frac{2k}{k+\kappa}A. \tag{2.6.4}$$

Now that the speed of the particle is different in the two regions we must use the probability current to calculate the reflection and transmission coefficients. The probability currents for incident, reflected, and transmitted waves are

$$J_{inc} = \frac{\hbar k}{m}|A|^2, \quad J_{ref} = \frac{\hbar k}{m}|B|^2, \qquad J_{tra} = \frac{\hbar\kappa}{m}|C|^2. \tag{2.6.5}$$

The reflection and transmission coefficients are

$$\begin{aligned} R = \frac{J_{ref}}{J_{inc}} &= \left|\frac{B}{A}\right|^2 = \left(\frac{k-\kappa}{k+\kappa}\right)^2 = \left(\frac{\sqrt{E}-\sqrt{E-V_0}}{\sqrt{E}+\sqrt{E-V_0}}\right)^2 \\ T = \frac{J_{tra}}{J_{inc}} &= \frac{\kappa}{k}\left|\frac{C}{A}\right|^2 = \frac{4k\kappa}{(k+\kappa)^2} = \frac{4\sqrt{E}\sqrt{E-V_0}}{(\sqrt{E}+\sqrt{E-V_0})^2} \end{aligned} \tag{2.6.6}$$

Observe that

$$R + T = 1, \tag{2.6.7}$$

consistent with conservation of probability. The coefficients and their sum as a function of energy are shown in the figure below.

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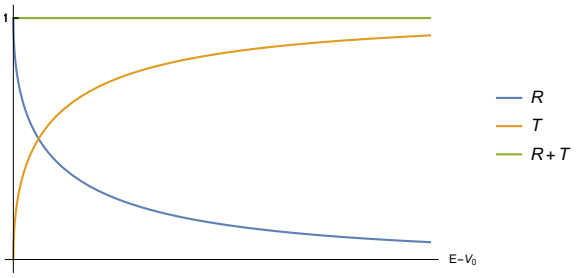


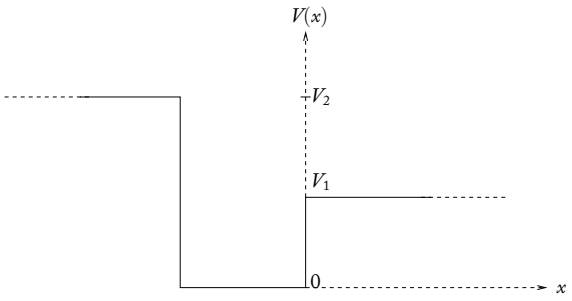
Figure 2.6.2: Reflection and transmission coefficients and their sum for a particle with energy above a step potential.

Question 2.6.1: Compare the reflection or transmission of the quantum particle discussed above with a classical particle.

Question 2.6.2: For $E < V_0$ in the step potential, (A). $B = A$, (B). $B = -A$, (C). $T = 1$, (D). $R = 1$,

Question 2.6.3: For $E << V_0$ in the step potential, (A). $B \to A$, (B). $B \to -A$, (C). $C \to A$, (D). $C \to -A$,

Question 2.6.4: Which of the following statements are true about the energies of a particle in the potential shown below?



(A). The energies are discrete for $0 < E < V_1$. (B). The energies are discrete for $V_1 < E < V_2$. (C). The energies are discrete for $E > V_2$. (D). The energies are continuous for $0 < E < V_1$. (E). The energies are continuous for $V_1 < E < V_2$. (F). The energies are continuous for $E > V_2$.

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2.7 The simple harmonic oscillator

Now we want to investigate a particle of mass m in the simple harmonic potential

$$V(x) = \frac{1}{2}kx^2 = \frac{1}{2}m\omega^2 x^2. \tag{2.7.1}$$

Because the potential is time-independent, we start with solving the time-independent Schrödinger equation,

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + \frac{1}{2}m\omega^2 x^2\psi(x) = E\psi(x). \tag{2.7.2}$$

We still need boundary conditions. Because the potential becomes infinite at $x = \pm\infty$, the particle cannot be found there, since it would need infinite energy. Alternatively, because the wavefunction needs to be normalizable, it has to vanish at $x = \pm\infty$. Therefore the boundary conditions are

$$\psi(-\infty) = 0, \qquad \psi(+\infty) = 0. \tag{2.7.3}$$

The wavefunction starts with $\psi(-\infty) = 0$ at $x = -\infty$, oscillates according to the Schrödinger equation in the interior, and falls back to $\psi(+\infty) = 0$ at $x = +\infty$. This equation is not easy to solve immediately. Therefore, we follow a number of steps.

Step 1: Simplify notation with change of variable

The first step is to rewrite the equation in simpler form without too many variables floating around. To do that, we make a change of variable of the coordinate x and define

$$\xi = \sqrt{\frac{m\omega}{\hbar}}x. \tag{2.7.4}$$

We then have $\frac{d}{dx} = \frac{d\xi}{dx}\frac{d}{d\xi} = \sqrt{\frac{m\omega}{\hbar}}\frac{d}{d\xi}$ and $\frac{d^2}{dx^2} = \frac{m\omega}{\hbar}\frac{d^2}{d\xi^2}$. The Schrödinger equation then takes the form

$$\frac{d^2\psi(\xi)}{d\xi^2} = (\xi^2 - K)\psi(\xi), \tag{2.7.5}$$

where

$$K \equiv \frac{2E}{\hbar\omega}. \tag{2.7.6}$$

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Step 2: Asymptotic behavior

If we have an approximate solution in some region of ξ, then we can write a solution ansatz that is a product of the approximate solution and an unknown function $H(\xi)$ that would allow to solve the general equation exactly. Hopefully, the equation for $H(\xi)$ is easier to solve. In particular, (2.7.5) is difficult because of the ξ^2 term. If we could have approximate solution that solves it at this order, then this term would be eliminated. Because the ξ^2 term dominates over the K term for large ξ, let us work in the large $\xi^2 >> K$ region, where the equation reduces to

$$\frac{d^2\psi(\xi)}{d\xi^2} \sim \xi^2\psi(\xi). \tag{2.7.7}$$

Now we can immediately write approximate solutions,

$$\psi(\xi) \sim e^{-\frac{1}{2}\xi^2}, \qquad \psi(\xi) \sim e^{\frac{1}{2}\xi^2}. \tag{2.7.8}$$

Because a solution that goes like $e^{\frac{1}{2}\xi^2}$ at large ξ cannot be normalizable, we discard it and keep only $\psi(\xi) \sim e^{-\frac{1}{2}\xi^2}$. Note that this gives $\psi''(\xi) \sim \xi^2 e^{-\frac{1}{2}\xi^2} - e^{-\frac{1}{2}\xi^2}$ which approximately equals $\xi^2 e^{-\frac{1}{2}\xi^2}$, since it dominates over $-e^{-\frac{1}{2}\xi^2}$ for large ξ, and thus the approximate equation is approximately solved. That is all well, since we are not seeking exact solutions at this point. We then write an ansatz solution,

$$\psi(\xi) = H(\xi)e^{-\frac{1}{2}\xi^2}; \tag{2.7.9}$$

we want to solve for $H(\xi)$ exactly and produce the exact $\psi(\xi)$. Differentiating this,

$$\frac{d^2\psi(\xi)}{d\xi^2} = H''(\xi)e^{-\frac{1}{2}\xi^2} - 2\xi H'(\xi)e^{-\frac{1}{2}\xi^2} + H(\xi)e^{-\frac{1}{2}\xi^2} + \xi^2 H(\xi)e^{-\frac{1}{2}\xi^2}. \tag{2.7.10}$$

Putting (2.7.9) and (2.7.10) in (2.7.5), we have

$$H''(\xi)e^{-\frac{1}{2}\xi^2} - 2\xi H'(\xi)e^{-\frac{1}{2}\xi^2} + H(\xi)e^{-\frac{1}{2}\xi^2} + \cancel{\xi^2 H(\xi)e^{-\frac{1}{2}\xi^2}} = \cancel{\xi^2 H(\xi)e^{-\frac{1}{2}\xi^2}} - KH(\xi)e^{-\frac{1}{2}\xi^2}. \tag{2.7.11}$$

Observe that the ξ^2 terms cancel out, since the approximate solution $e^{-\frac{1}{2}\xi^2}$ solves the Schrödinger equation at leading order in large ξ.

Thus, the Schrödinger equation reduces to an equation for $H(\xi)$.

$$H''(\xi) - 2\xi H'(\xi) + (K-1)H(\xi) = 0. \tag{2.7.12}$$

This equation is called the Hermite equation.

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Step 3: Method of power series

It is known in the theory of differential equations that if the equation is written with the coefficient of the highest order derivative set to one as in here, then the equation could be singular only at the singular points of the coefficients of the lower order derivative terms. The coefficients we have in (2.7.12) have no singularity and, therefore, $H(\xi)$ can have no singularities at finite ξ and a power series solution will have an infinite radius of convergence. Thus, we can write a power series solution of the form

$$H(\xi) = \sum_{n=0}^{\infty} a_n \xi^n. \tag{2.7.13}$$

The idea is then to rewrite (2.7.12) as $\sum[\ldots]_n \xi^n = 0$, with constant coefficients $[\ldots]_n$, which holds for all ξ only if each constant coefficient vanishes; i.e., $[\ldots]_n = 0$ for each n. Therefore, we seek to write each term in (2.7.12) as $\sum[\ldots]_n \xi^n = 0$ so that we can easily combine the terms together. First, $H'(\xi) = \sum_{n=0}^{\infty} n a_n \xi^{n-1}$. Because what we have in (2.7.12) is $-2\xi H'(\xi) = -2\sum_{n=0}^{\infty} n a_n \xi^n$, which has the desired form, we keep it as is. Next, taking the second derivative, $H''(\xi) = \sum_{n=0}^{\infty} n(n-1) a_n \xi^{n-2}$. Now this has ξ^{n-2} while we seek ξ^n so that we can combine the coefficients with the other terms. But the first two terms for $n = 0$ and $n = 1$ are zero, and we can start the sum with $n = 2$, $H''(\xi) = \sum_{n=2}^{\infty} n(n-1) a_n \xi^{n-2}$. Next we shift $n \to n+2$ and the sum starts at $n = 0$ as in the others, which we do by rewriting it as $H''(\xi) = \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} \xi^n$, and we have expanded it in terms of ξ^n too. Putting these in (2.7.12), we have

$$\sum_{n=0}^{\infty} \left[(n+2)(n+1) a_{n+2} - (2n+1-K) a_n\right] \xi^n = 0. \tag{2.7.14}$$

This holds for all ξ only if each and every coefficient of ξ^n is identically zero, which gives the recursion relation

$$a_{n+2} = \frac{2n+1-K}{(n+2)(n+1)} a_n \tag{2.7.15}$$

Observe that there are only two independent coefficients, since starting with a_0, we recursively produce a_2, a_4, etc, which produces even solutions. If we start with a_1, we recursively obtain a_3, a_5, etc, which produces odd solutions.

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Step 4: Infinite series produces a diverging function

For large n, we have

$$\frac{a_{n+2}}{a_n} \to \frac{2}{n} \tag{2.7.16}$$

But this is similar to the behavior in the series expansion of $\xi^m e^{\xi^2}$ for any m, which goes like e^{ξ^2} for large ξ. In other words, if we keep all infinite series of terms, then we would have $H(\xi) \sim e^{\xi^2}$ and $\psi(\xi) = H(\xi)e^{-\frac{1}{2}\xi^2} \sim e^{\frac{1}{2}\xi^2}$, and the wavefunction would blow up at large ξ and would not be normalizable.

Step 5: Truncate series

We need to truncate the sum so that $H(\xi)$ would at most be a polynomial of some finite degree n, and then $\psi(\xi) = H(\xi)e^{-\frac{1}{2}\xi^2}$ would be finite, since $e^{-\frac{1}{2}\xi^2}$ decays much stronger than any polynomial of finite degree at large ξ. We truncate the series at n by setting $a_{n+2} = 0$ while $a_n \neq 0$, which is achieved when $2n + 1 - K = 0$, or using the definition of K,

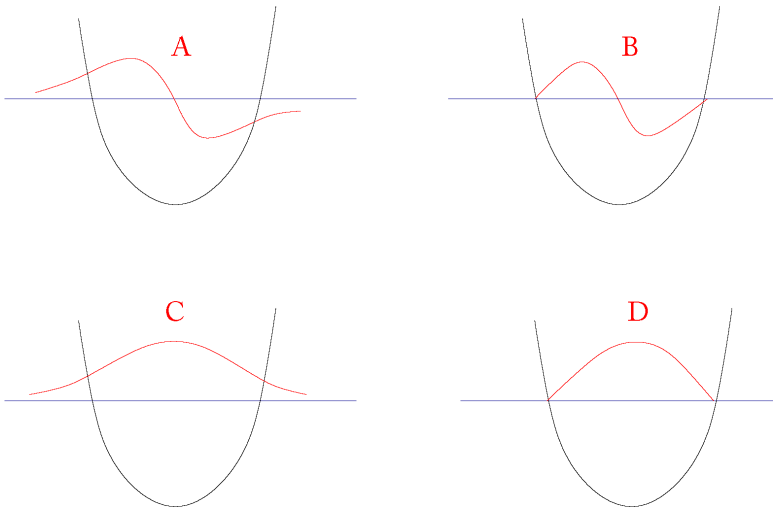
$$E_n = (n + \frac{1}{2})\hbar\omega, \tag{2.7.17}$$

where we could choose to truncate the series at $n = 0$, $n = 1$, $n = 2$, etc, which produce polynomials of the same degree. These energies are the energy levels of the simple harmonic oscillator, and the corresponding polynomials are the Hermite polynomials (up to irrelevant overall constant factors).

2.7.1 Simple harmonic oscillator wavefunctions

Question 2.7.1: Which one of the following best resembles the first excited state of the simple harmonic oscillator?

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Let us start with truncating the series at $n = 0$ such that $a_0 \neq 0$ while $a_2 = 0$. Because $a_2 = \frac{2n+1-K}{(n+2)(n+1)} a_n|_{n=0} = \frac{1-K}{2} a_0$, we set $1 - K = 0$ which gives

$$E_0 = \frac{1}{2}\hbar\omega. \tag{2.7.18}$$

We also have $H_0(\xi) = a_0$, and the wavefunction is $\psi_0(\xi) = H_0(\xi)e^{-\frac{1}{2}\xi^2} = a_0 e^{-\frac{1}{2}\xi^2} = a_0 e^{-\frac{m\omega}{2\hbar}x^2}$. Normalizing, the ground state wavefunction is

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega}{2\hbar}x^2}. \tag{2.7.19}$$

To construct the first excited state, we truncate the series at $n = 1$ such that $a_3 = 0$ while $a_1 \neq 0$. Therefore, $a_3 = \frac{2n+1-K}{(n+2)(n+1)} a_n|_{n=1} = \frac{3-K}{6} a_1 = 0$, or $3 - K = 0$, which gives

$$E_0 = \frac{3}{2}\hbar\omega. \tag{2.7.20}$$

Furthermore, $H_1(\xi) = a_1\xi$, and the wavefunction is $\psi_1(\xi) = H_1(\xi)e^{-\frac{1}{2}\xi^2} = a_1\xi e^{-\frac{1}{2}\xi^2} = a_1\sqrt{\frac{m\omega}{\hbar}} x e^{-\frac{m\omega}{2\hbar}x^2}$. Normalizing, the first excited state wavefunction is

$$\psi_1(x) = \sqrt{2}\left(\frac{m^3\omega^3}{\pi\hbar^3}\right)^{1/4} x e^{-\frac{m\omega}{2\hbar}x^2}. \tag{2.7.21}$$

The polynomials $H_n(\xi)$ are called the Hermite polynomials (up to an irrelevant overall constant factor).

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Now we can continue to construct any of the stationary wavefunctions. Suppose we want to construct $\psi_n x$. First, we find K by setting $a_{n+2} = 0$ while $a_n \neq 0$. We then use this K to determine a_2, a_4, ..., a_n in terms of a_0 for even n, or a_3, a_5, ..., a_n in terms of a_1 for odd n. Note that the coefficients are generally different for different states, since the coefficients depend on K (or energy).

The wavefunction $\psi_n(x)$ for the nth state could be conveniently written as

$$\psi_n(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \frac{1}{\sqrt{2^n n!}} H_n(\xi) e^{-\frac{1}{2}\xi^2}, \tag{2.7.22}$$

where the Hermite polynomials $H_n(\xi)$ are given by the Rodrigues formula

$$H_n(\xi) = (-1)^n e^{\xi^2} \left(\frac{d}{d\xi}\right)^n e^{-\xi^2} \tag{2.7.23}$$

It can be shown that these wavefunctions form a complete orthonormal set for square-integrable wavefunctions of the harmonic oscillator.

The first few wavefunctions and the corresponding energies are shown in the schematic figure below.

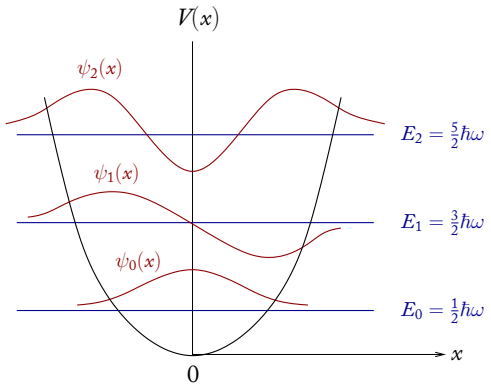


Figure 2.7.1: Wavefunctions and energies for a particle in the simple harmonic oscillator potential.

Question 2.7.2: Discuss why the stationary states for large n are peaked near the boundary of the simple harmonic oscillator as shown in the figure below, in contrast to the low lying states.

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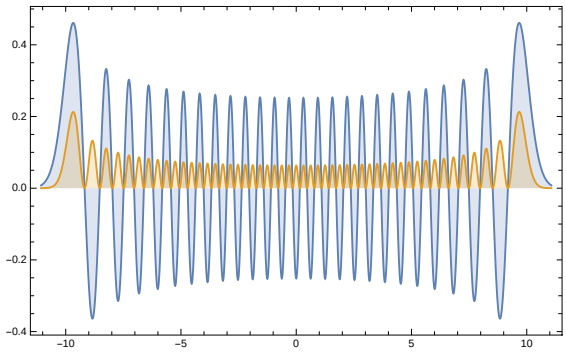


Figure 2.7.2: Wavefunction (blue) and probability density (orange) of the simple harmonic oscillator for large n, $n = 50$ taken here.

Exercise 2.7.1: Show the following Gaussian integrals.

1. $\int_{-\infty}^{\infty} e^{-ax^2}\,dx = \sqrt{\frac{\pi}{a}}$ and $\int_{0}^{\infty} e^{-ax^2}\,dx = \frac{1}{2}\sqrt{\frac{\pi}{a}}$.
2. $\int_{-\infty}^{\infty} x^{2n+1} e^{-ax^2}\,dx = 0$ for $n = 0, 1, 2, \ldots$
3. $\int_{-\infty}^{\infty} x^{2n} e^{-ax^2}\,dx = (-1)^n \frac{d^n}{da^n}\sqrt{\frac{\pi}{a}}$ for $n = 0, 1, 2, \ldots$
4. $\int_{-\infty}^{\infty} e^{-ax^2+bx}\,dx = \sqrt{\frac{\pi}{a}} e^{\frac{b^2}{4a}}$.
5. $\int_{-\infty}^{\infty} x^n e^{-ax^2+bx}\,dx = \frac{d^n}{db^n}\sqrt{\frac{\pi}{a}} e^{\frac{b^2}{4a}}$.

Exercise 2.7.2:

1. Starting with $\psi(\xi) = H(\xi)e^{-\frac{1}{2}\xi^2}$, $H(\xi) = \sum_{n=0}^{\infty} a_n \xi^n$, and the recursion relation $a_{n+2} = \frac{2n+1-K}{(n+2)(n+1)} a_n$ with $\xi \equiv \sqrt{\frac{m\omega}{\hbar}}\, x$ and $K \equiv \frac{2E}{\hbar\omega}$, determine the energies and the corresponding wavefunctions for the second ($n = 2$) and the third ($n = 3$) excited states of a particle of mass m in a simple harmonic oscillator with potential $V(x) = \frac{1}{2}m\omega^2 x^2$.
2. Starting with the Rodrigues formula for $H(\xi)$, $H_n(\xi) = (-1)^n e^{\xi^2}\left(\frac{d}{d\xi}\right)^n e^{-\xi^2}$, determine the wavefunctions for the second ($n = 2$) and the third ($n = 3$) excited states, check that the results are the same as in part (1) (up to irrelevant overall constant factors).
3. Determine the uncertainties for position Δx and momentum Δp for a particle in the ground state of the simple harmonic oscillator, and check whether the uncertainty principle holds.

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4. The particle is in state $\psi(x,0) = \frac{1}{\sqrt{2}}(\psi_0(x) + \psi_1(x))$ of the simple harmonic oscillator at time $t = 0$.

   (a) Write down the wavefunction $\psi(x,t)$ at time t. Set $\hbar = m = \omega = 1$ and use Mathematica to plot $|\psi(x,t)|^2$ at $t = 0$, $t = \frac{\pi}{2}$, $t = \pi$, $t = \frac{3\pi}{2}$, and $t = 2\pi$. (5 points)

   (b) Determine $\langle \hat{x} \rangle$ at time t, and find the amplitude and frequency of oscillation of $\langle \hat{x} \rangle$.

   (c) Determine $\langle \hat{p} \rangle$ at time t.

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